

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0706

COURSE NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I C KARTHIK REDDY (192425157) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

C KARTHIK REDDY

Question 1

Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial $x^3 + 1$.

- (i) Use polynomial long division to determine the message that should be transmitted.
- (ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

Answer

Introduction

Cyclic Redundancy Check (CRC) is one of the most widely used error detection techniques in computer networks. It is used in protocols such as Ethernet, Wi-Fi, USB, and many other communication systems to detect errors that occur during data transmission.

The sender appends a CRC remainder to the original message before transmission. At the receiver side, the received message is divided again by the same generator polynomial. If the remainder is zero, the data is accepted. Otherwise, the receiver concludes that an error has occurred.

Given Data

Original Message

11100011

Generator Polynomial

$G(x) = x^3 + 1$

Binary Generator

1001

Degree of Generator Polynomial

3

Therefore, append **3 zeros** to the message before performing CRC division.

Step 1: Append Zeros

Original Message

11100011

Appending three zeros

11100011000

This is called the dividend.

Step 2: Perform Binary Polynomial Division

Generator

1001

Dividend

11100011000

Binary XOR division is performed.

```
      1001 )11100011000
            1110
            1001
            ----
             0111
             1110
             1001
             ----
              0110
              1100
              1001
              ----
               0101
               1011
               1001
               ----
                0010
                1001
                1001
                ----
                 0000
```

Remainder = 001

Hence,

CRC Remainder

001

Step 3: Form the Transmitted Message

The CRC bits are appended to the original message.

Original Message

11100011

CRC

001

Therefore,

Transmitted Message

11100011001

This is the data transmitted over the communication channel.

Answer for Part (i)

The transmitted message after adding the CRC bits is

11100011001

Part (ii)

Suppose noise changes the leftmost bit.

Original transmitted message

11100011001

Received message

01100011001

The receiver again performs CRC division using the same generator polynomial.

Generator

1001

Received Data

01100011001

After binary division,

Remainder $\neq$ 000

(Non-zero remainder)

Since the remainder is non-zero, the receiver concludes that an error has occurred during transmission.

How CRC Detects Errors

CRC works because both sender and receiver use the same generator polynomial.

- Sender divides the message by the generator polynomial.
- The remainder is appended to the message.
- Receiver divides the received message by the same generator.
- If the remainder is zero, the message is considered correct.
- If the remainder is non-zero, an error has occurred.

Even if only one bit changes during transmission, the CRC remainder changes, allowing the receiver to detect the error.

Advantages of CRC

1. Detects single-bit errors.
2. Detects burst errors effectively.
3. Very reliable compared to simple parity checking.
4. Fast and easy to implement in hardware and software.
5. Widely used in Ethernet, Wi-Fi, USB, storage devices, and communication systems.

Applications of CRC

- Ethernet Networks
- Wi-Fi Networks
- Satellite Communication
- Mobile Networks
- USB Communication
- Digital Television
- Hard Disk Storage

- Data Communication Protocols

Result

Parameter	Value
Original Message	11100011
Generator Polynomial	$x^3 + 1$
Binary Generator	1001
CRC Remainder	001
Transmitted Message	11100011001
Error Introduced	Leftmost bit inverted
Receiver Remainder	Non-zero
Final Decision	Error detected and frame rejected

QUESTION 2(a)

A network with a bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute, with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Answer

Introduction

Throughput is the actual amount of data successfully transmitted over a network in a given amount of time. Although a network may have a high bandwidth, the actual throughput is usually lower because of delays, congestion, protocol overhead, retransmissions, and network traffic.

Bandwidth represents the maximum transmission capacity of the communication channel, whereas throughput indicates the real performance of the network.

Given Data

- Bandwidth = **10 Mbps**
- Number of frames transmitted per minute = **12,000**
- Size of each frame = **10,000 bits**

Formula

$$\text{Throughput} = \frac{\text{Total Bits Transmitted}}{\text{Time}}$$

Step 1: Calculate Total Bits Transmitted in One Minute

$$12000 \times 10000 = 120000000 \text{ bits}$$

Therefore,

Total bits transmitted in one minute = 120,000,000 bits

Step 2: Convert into Bits per Second

Since,

$$\begin{aligned} 1 \text{ minute} &= 60 \text{ seconds} \\ \text{Throughput} &= \frac{120000000}{60} \\ &= 2000000 \text{ bits/sec} \end{aligned}$$

Step 3: Convert into Mbps

Since,

$$\begin{aligned} 1 \text{ Mbps} &= 1000000 \text{ bps} \\ \frac{2000000}{1000000} &= 2 \text{ Mbps} \end{aligned}$$

Answer

Throughput = 2 Mbps

Interpretation

The network is capable of supporting **10 Mbps**, but only **2 Mbps** is being utilized effectively.

Efficiency

$$\frac{2}{10} \times 100 = 20\%$$

Thus, only **20% of the available bandwidth** is utilized.

QUESTION 2(b)

A network has a bandwidth of 3000 Hz (300–3300 Hz) assigned for data communication. The signal-to-noise ratio is 3162. Find the channel capacity.

Answer

Introduction

The maximum amount of data that can be transmitted over a communication channel without

errors is called the **channel capacity**.

Claude Shannon developed the Shannon Capacity Theorem, which determines the theoretical maximum data rate of a noisy communication channel.

Given Data

Bandwidth

$$B = 3000 \text{ Hz}$$

Signal-to-Noise Ratio

$$S/N = 3162$$

Formula

Shannon Capacity Formula

$$C = B \log_2(1 + S/N)$$

where

- C = Channel Capacity
- B = Bandwidth
- S/N = Signal-to-Noise Ratio

Step 1: Substitute Values

$$C = 3000 \log_2(3163)$$

Step 2: Calculate Logarithm

$$\log_2(3163) \approx 11.63$$

Step 3: Calculate Capacity

$$\begin{aligned} C &= 3000 \times 11.63 \\ &= 34890 \text{ bps} \end{aligned}$$

Final Answer

Channel Capacity ≈ 34.9 kbps

Interpretation

This means the communication channel can theoretically transmit approximately **34.9 kbps**

without errors under the given bandwidth and noise conditions.

Increasing either bandwidth or signal-to-noise ratio increases channel capacity.

Applications

- Telephone networks
- DSL communication
- Wireless communication
- Fiber-optic systems
- Satellite communication

QUESTION 3

A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the Hamming (12,8) error-correcting code.

Answer

Introduction

Wireless communication is widely used in healthcare systems for real-time patient monitoring. However, wireless channels are affected by electromagnetic interference, which may introduce bit errors during transmission.

To ensure that patient data is received accurately without retransmission delays, hospitals use **Hamming codes**, which can detect and correct single-bit errors.

Theory of Hamming (12,8) Code

Hamming (12,8) is an error-correcting code in which:

- Total bits = 12
- Data bits = 8
- Parity bits = 4

The parity bits are placed at positions:

- P1
- P2
- P4

- P8

The remaining positions store the data bits.

Working Process

Step 1

Patient monitoring device records:

- Heart Rate
- Blood Pressure

Step 2

The information is converted into an 8-bit binary message.

Example:

10110010

Step 3

Four parity bits are generated.

The transmitted frame becomes

P1 P2 D1 P4 D2 D3 D4 P8 D5 D6 D7 D8

Total = 12 bits

Step 4

The encoded data is transmitted through the wireless channel.

Step 5

Suppose one bit changes due to electromagnetic interference.

Example

101100101011

↓

101100111011

Step 6

At the receiver, parity bits are checked.

The syndrome value identifies the exact position of the incorrect bit.

Step 7

The receiver flips that bit back to its correct value.

The corrected data is delivered immediately.

Advantages

- Detects single-bit errors
- Corrects single-bit errors automatically
- No retransmission required
- Fast communication
- High reliability
- Suitable for medical monitoring systems

Applications

- Hospital monitoring systems
- Computer memory (ECC RAM)
- Satellite communication
- Wireless sensor networks
- Space communication
- Networking devices

Result

The Hamming (12,8) code detects and corrects one-bit transmission errors automatically, ensuring accurate and reliable delivery of patient health data without waiting for retransmission.

Question 4(a) IPv4 Subnetting

A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments.

Using the given subnet mask:

1. Calculate the total number of subnets.
2. Calculate the number of usable hosts per subnet.
3. Determine the subnet to which **172.16.10.78** belongs.
4. Find the network address and broadcast address.
5. Determine whether **172.16.10.95** belongs to the same subnet.

Answer

Introduction

Subnetting is the process of dividing a large network into smaller logical networks called **subnets**. It improves network performance, enhances security, and reduces unnecessary broadcast traffic.

IPv4 addresses consist of **32 bits** divided into:

- Network ID
- Host ID

Subnetting borrows bits from the host portion to create multiple smaller networks.

Given Data

- Network Address = **172.16.10.0/24**
- Subnet Mask = **/27**

Step 1: Determine Borrowed Bits

Original Prefix = **/24**

New Prefix = **/27**

Borrowed Bits

$$27 - 24 = 3$$

Therefore,

Borrowed bits = 3

Step 2: Calculate Number of Subnets

Formula

$$2^{\text{Borrowed Bits}}$$
$$2^3 = 8$$

Answer

Total Subnets = 8

Step 3: Calculate Hosts per Subnet

Host Bits

$$32 - 27 = 5$$

Formula

$$2^{HostBits} - 2$$

$$2^5 - 2$$

$$32 - 2 = 30$$

Answer

Usable Hosts per Subnet = 30

Step 4: Determine Subnet Ranges

Each subnet contains **32 addresses**.

Subnet	Network Address	Host Range	Broadcast Address
1	172.16.10.0	1–30	31
2	172.16.10.32	33–62	63
3	172.16.10.64	65–94	95
4	172.16.10.96	97–126	127
5	172.16.10.128	129–158	159
6	172.16.10.160	161–190	191
7	172.16.10.192	193–222	223
8	172.16.10.224	225–254	255

Step 5: Find the Subnet of 172.16.10.78

IP Address

172.16.10.78

78 lies between **64 and 95**

Therefore,

Subnet

172.16.10.64/27

Network Address

172.16.10.64

Broadcast Address

172.16.10.95

Usable Host Range

172.16.10.65

to

172.16.10.94

Step 6: Check IP Address 172.16.10.95

172.16.10.95 is the **broadcast address** of the subnet.

It belongs to the same subnet but **cannot be assigned to a host**.

Final Answers

- Number of Subnets = **8**
- Hosts per Subnet = **30**
- Network Address = **172.16.10.64**
- Broadcast Address = **172.16.10.95**
- 172.16.10.95 is the **broadcast address**, not a valid host IP.

Applications of Subnetting

- Reduces network congestion.
- Improves network performance.
- Enhances security.
- Simplifies network management.
- Efficient use of IP addresses.

Question 4(b) IPv6 Addressing

Given the IPv6 address

2001:0db8:0000:0000:0000:ff00:0042:8329

Perform the following:

1. Compress the address.
2. Expand it again.
3. Find the network prefix using **/64**.
4. Calculate the total number of host addresses.

Answer

Introduction

IPv6 is the latest version of the Internet Protocol. It uses **128-bit addresses**, providing a much larger address space than IPv4. IPv6 also supports efficient routing, auto-configuration, and improved security.

Given Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the Address

Rules:

- Remove leading zeros.
- Replace one continuous sequence of zero groups with ::.

Compressed Address

2001:db8::ff00:42:8329

Step 2: Expand the Address

Expanded Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine Network Prefix

Prefix Length

/64

The first 64 bits represent the network.

Network Prefix

2001:db8:0000:0000::/64

Step 4: Calculate Host Addresses

Remaining Bits

$$128 - 64 = 64$$

Number of Host Addresses

$$\begin{aligned} &2^{64} \\ &= 18,446,744,073,709,551,616 \end{aligned}$$

Answer

- Compressed Address

2001:db8::ff00:42:8329

- Expanded Address

2001:0db8:0000:0000:0000:ff00:0042:8329

- Network Prefix

2001:db8:0000:0000::/64

- Number of Host Addresses

2^{64}

=

18,446,744,073,709,551,616

Advantages of IPv6

- Huge address space.
- Faster routing.
- Better security (IPSec support).
- Stateless auto-configuration.
- Eliminates NAT.
- Improved Quality of Service (QoS).

Question 5

University Online Examination Scenario

Answer

Introduction

Flow control is a mechanism that regulates the rate of data transmission between a sender and a receiver. It ensures that a fast sender does not overwhelm a slower receiver. In modern computer networks, flow control is implemented using protocols such as **TCP** to guarantee reliable communication.

Scenario Analysis

The university is conducting an online examination for **8,000 students** simultaneously.

- Server speed = **10 Gbps**

- Student connections = **100 Mbps, 20 Mbps, and 5–10 Mbps**

Because the server transmits data much faster than many students can receive it, communication problems occur.

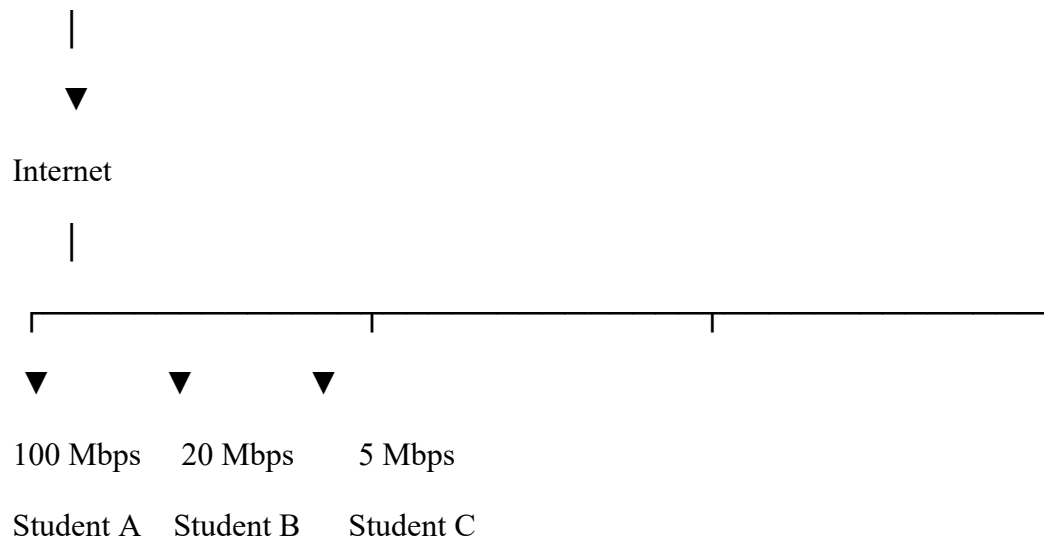
Primary Networking Problem

The primary networking problem is **Flow Control Failure (Sender–Receiver Speed Mismatch)**.

How the Problem Occurs

University Server

(10 Gbps)



The server continues sending packets at a very high speed. Students with slower connections cannot process the packets quickly enough.

Effects of the Speed Mismatch

1. Receiver Buffer Overflow

- Packets arrive faster than they can be processed.
- Receiver memory becomes full.

2. Packet Loss

- Excess packets are dropped.

3. Retransmissions

- Lost packets are sent again, increasing network traffic.

4. Delayed Acknowledgements

- ACK packets are delayed due to congestion.

5. Higher Latency

- Students experience delays while downloading question papers.

6. Portal Freezing

- Browsers appear to hang because required data has not arrived completely.

7. Network Congestion

- Retransmissions consume additional bandwidth.

Impact on the Examination

- Late receipt of question papers.
- Unequal start times for students.
- Increased server load.
- Poor user experience.
- Reduced fairness and reliability of the examination process.

Solution

The university should implement **TCP Flow Control** using a **Sliding Window** mechanism.

Additional measures include:

- Congestion control.
- Load balancing.
- Content Delivery Networks (CDNs).
- Adequate receiver buffer sizes.
- Multiple examination servers.

Advantages of Flow Control

- Prevents buffer overflow.
- Reduces packet loss.
- Improves throughput.
- Minimizes retransmissions.
- Ensures reliable communication.
- Provides a smooth user experience.